

INTRO TO QUARTO

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12 Nov 2026

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Why reproducibility matters

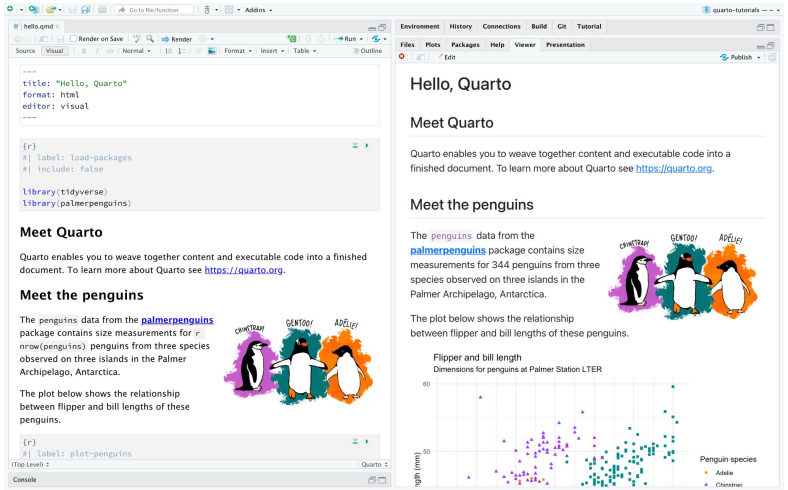
- Your analysis should rerun from raw data to final figure
- Supervisors, examiners, and collaborators expect it
- Journals increasingly require data and code
- Future you is a collaborator too

The problem with copy-paste

- Excel + R + Word copies drift apart
- Update the data: every table and figure must be redone by hand
- A rerun breaks everything

What is Quarto?

- Quarto is the next-generation successor of R Markdown
- Works with R, Python, Julia and Observable
- Built into RStudio 2022.07 or later
- Website: quarto.org
- Files end with .qmd



Anatomy of a Quarto document

1. YAML header: title, author, format
2. Markdown text: headings, bold, lists
3. R chunks: code that runs

```
---
title: "My report"
format: html
editor: visual
---
```

- More advanced YAML

```
---
title: "Quarterly Sales Report"
author: "Jane Doe"
date: "2026-11-11"
format:
     html:
          toc: true
          theme: cosmo
execute:
     echo: false
---
```

Chunk options

- echo: show or hide the code
- eval: run the code or not
- warning / message: show or hide
- Written with #| at the top of the chunk
- #| echo: false
- #| label: fig-age
- Give chunks labels

````
```{r}
#| label: fig-penguins
#| fig-cap: "Dimensions of penguins across three species."
#| warning: false
library(tidyverse, quietly = TRUE)
library(palmerpenguins)
penguins |>
  ggplot(aes(x = flipper_length_mm, y = bill_length_mm)) +
  geom_point(aes(color = species)) +
  scale_color_manual(
    values = c("darkorange", "purple", "cyan4")) +
  theme_minimal()
```
````

Rendering

- Click the Render button or run quarto render
- HTML: easiest, opens in any browser
- PDF: needs LaTeX (tinytex)
- Word (.docx): good for if others want to comment
- Rendering runs the whole document from top to bottom

Example

SOURCE FILE: hello.qmd

```
---
title: "Hello, Penguins"
format: html
execute:
  echo: false
---

## Meet the penguins

The `penguins` data contain
from three islands in the P

The three species of penguin
distributions of physical dimensions (@fig-penguins).

```{r}
#| label: fig-penguins
#| fig-cap: "Dimensions of penguins a
#| warning: false
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  ggplot(aes(x = flipper_length_mm, y = bill_length_mm)) +
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  scale_color_manual(
    values = c("darkorange", "purple", "cyan4")) +
```

Set format(s) and options
Use YAML Syntax

Write with **Markdown**
RStudio: Help > Markdown Quick Reference
Use **Visual Editor**

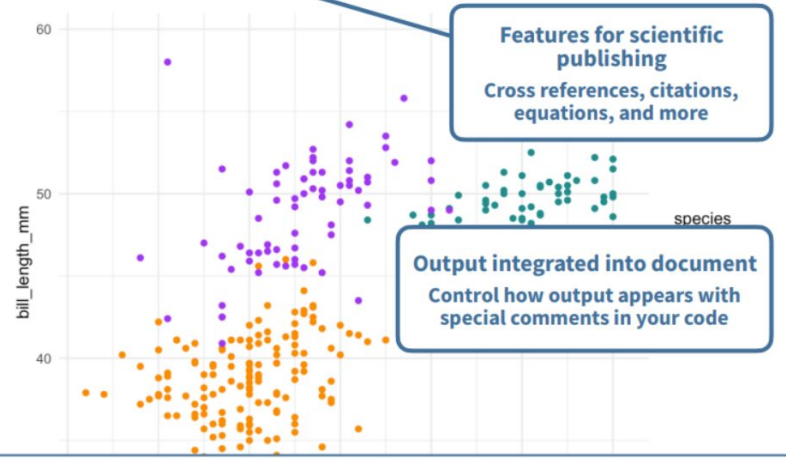
Include code
R, Python, Julia, Observable, or any language with a Jupyter kernel

RENDERED OUTPUT: hello.html

Hello, Penguins

Meet the penguins

The three species of penguins have quite distinct distributions of physical dimensions (Figure 1).



Features for scientific publishing
Cross references, citations, equations, and more

Output integrated into document
Control how output appears with special comments in your code
```

Live demo plan

1. Create a new .qmd file
2. Add a YAML header
3. Insert an R chunk
4. Render to html and Word (docx)

Further readings

- https://rstudio.github.io/cheatsheets/html/quarto.html
- https://rstudio.github.io/cheatsheets/quarto.pdf



Any question?